Smart Attendance System - Project Documentation

1. Project Overview

The **Smart Attendance System** is a modern, full-stack web application designed to automate student attendance tracking using Artificial Intelligence. By leveraging facial recognition technology, the system eliminates traditional manual methods, providing a secure, efficient, and data-driven approach to classroom management.

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2. Key Features

- **AI Face Recognition**: Automated attendance capture using `face-api.js` for real-time student identification.
- **Admin Dashboard**: A comprehensive control center with live statistics and status updates.
- **Student Management**: Full-featured directory to register students, manage records, and reset credentials.
- **Performance Analytics**: Visualized correlation between physical attendance and academic results.
- **Real-time Statistics**: Instant tracking of attendance percentages and top-performing students.
- **Secure Authentication**: Role-based access control (Admin/Student) with JWT-based security.

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3. Technology Stack

Frontend

- **Framework**: React 18 (Vite-powered)
- **Styling**: Tailwind CSS
- **AI/ML**: Face-api.js & TensorFlow.js
- **Charts**: Recharts
- **Icons**: Lucide-react

Backend

- **Runtime**: Node.js
- **Framework**: Express.js
- **Database**: SQLite (via `sql.js` & `sqlite3`)
- **Security**: Bcrypt.js & JSON Web Tokens (JWT)
- **Validation**: Zod

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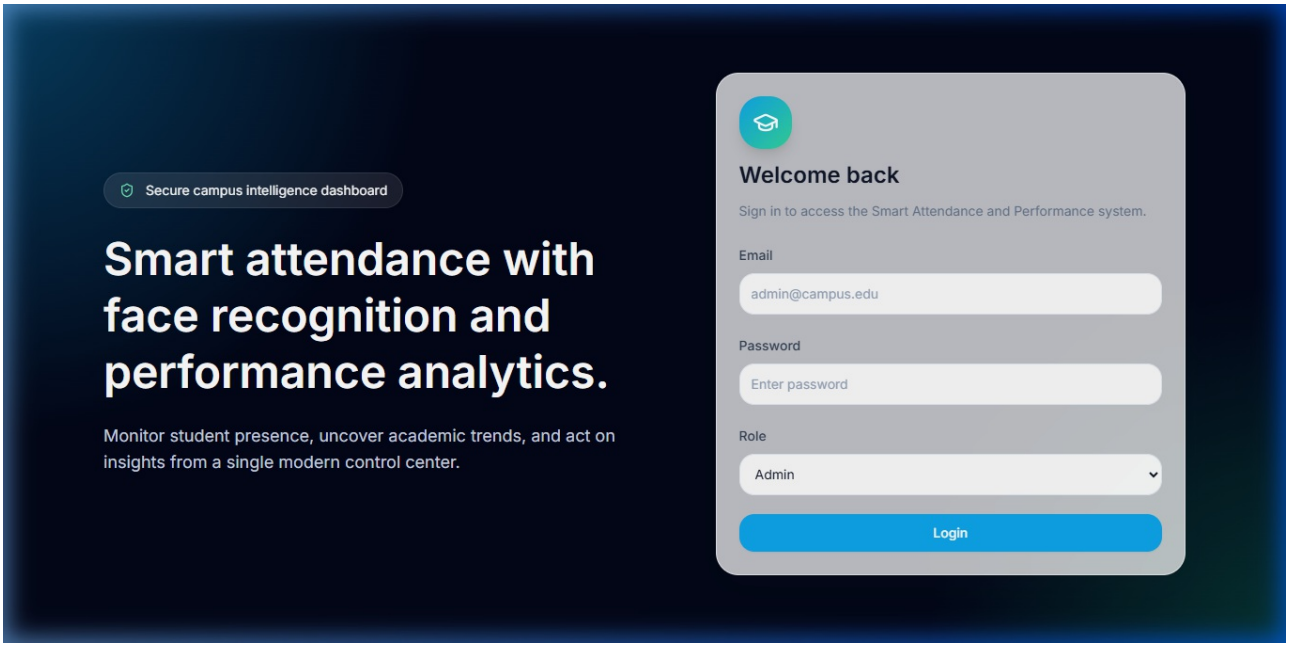
4. Visual Walkthrough

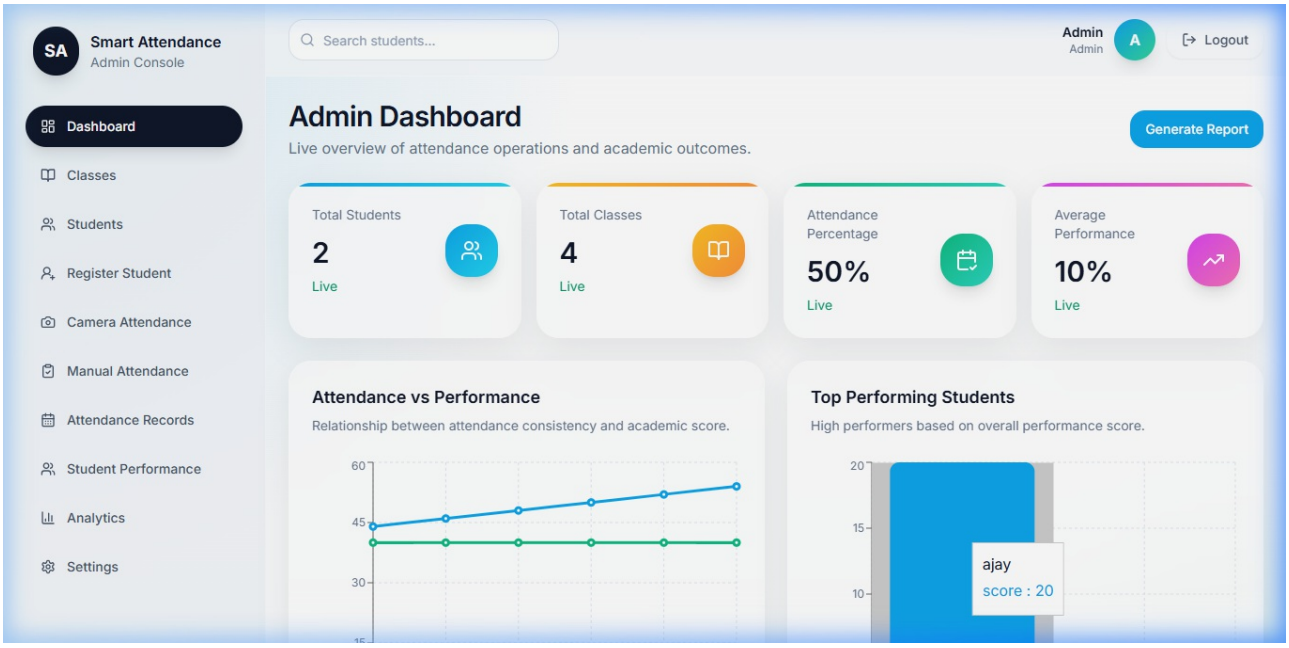
4.1 Landing & Login

The application features a sleek, dark-themed landing page. Administrators can log in through a secure portal to access the management console.



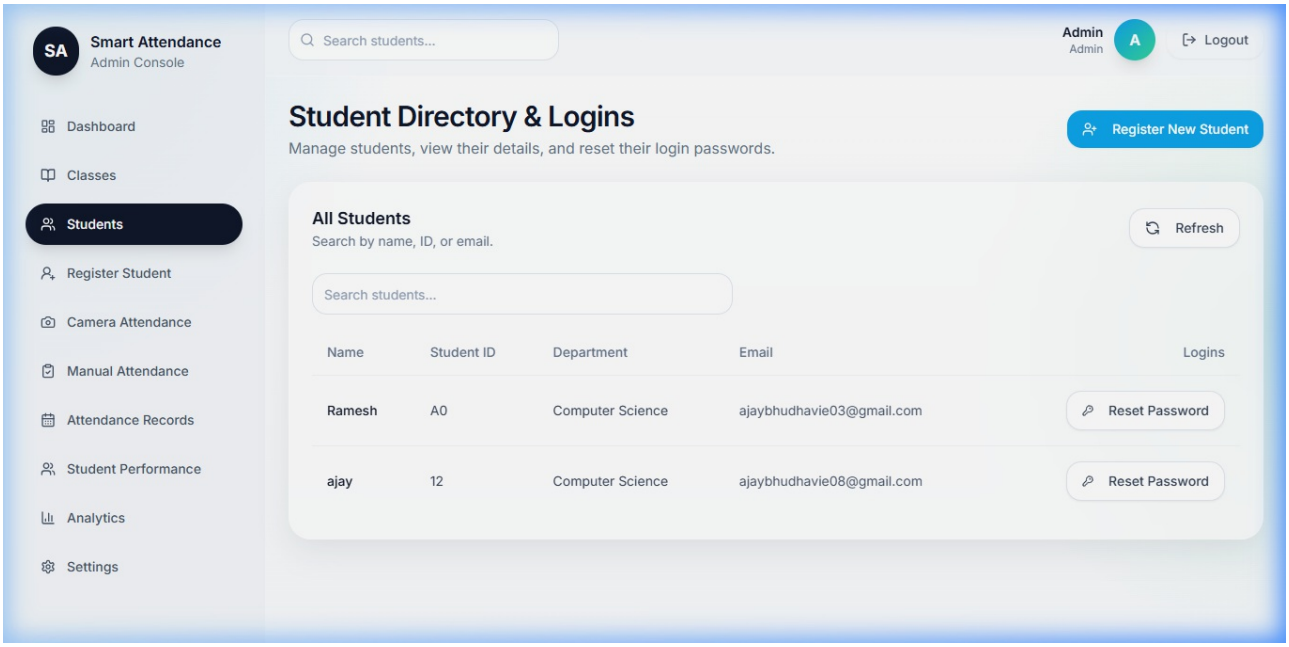
4.2 Administrative Dashboard

The dashboard provides a high-level overview of the campus operations, including total students, active classes, and average attendance performance.



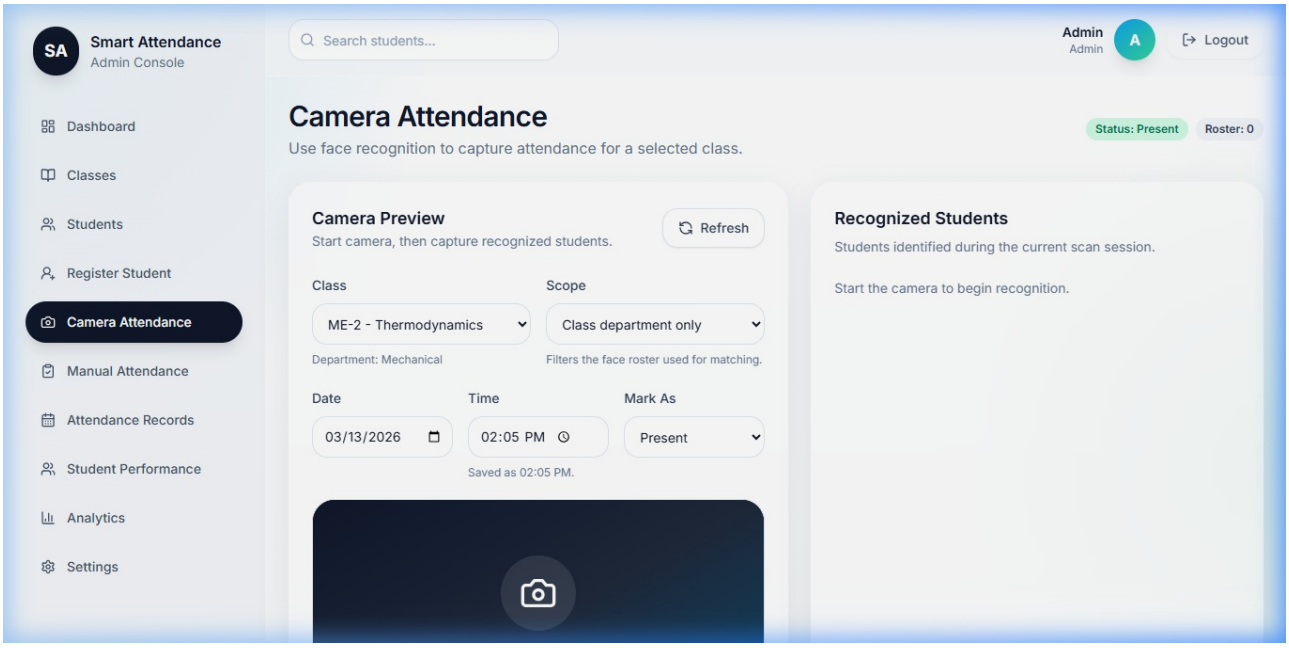
4.3 Student Directory

Administrators can manage the entire student body, search by name or ID, and handle administrative tasks like password resets.



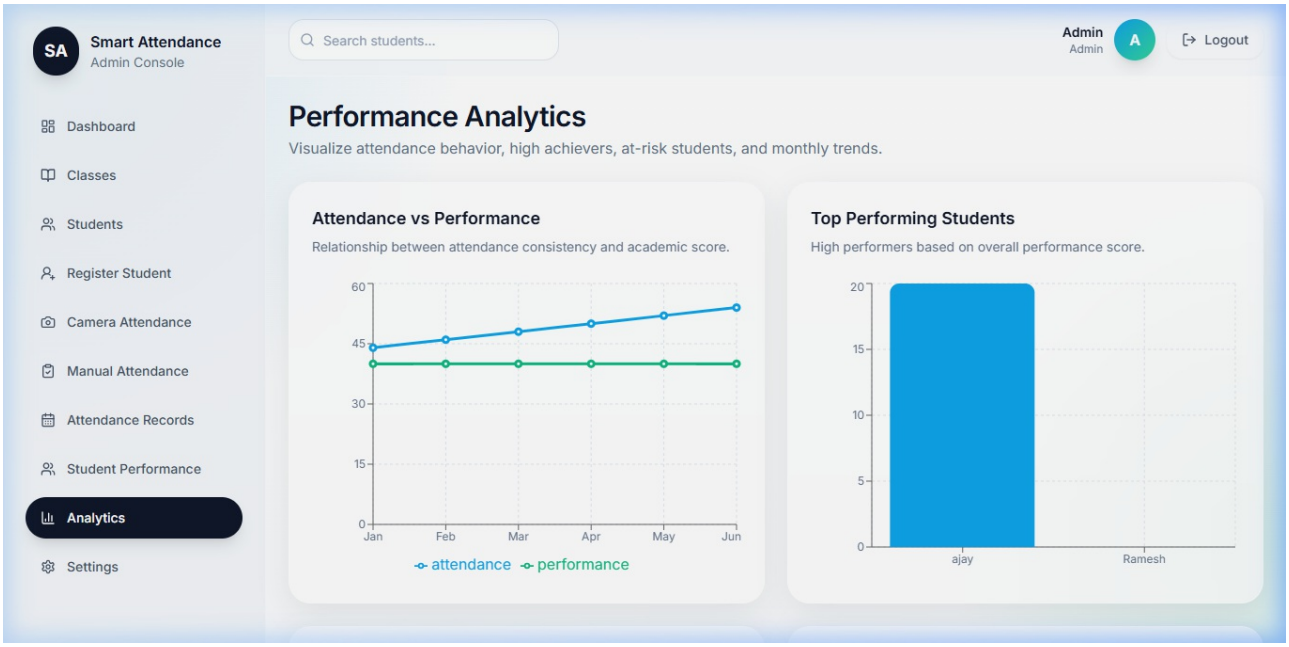
4.4 Camera-Based Attendance

The core feature of the system. It uses the device's camera to recognize students' faces and automatically marks them present for the selected class session.



4.5 Performance Analytics

This module provides deep insights into student behavior, allowing educators to identify at-risk students and celebrate high achievers.



5. System Architecture

The project follows a decoupled client-server architecture:

- **Client**: Handles the UI and heavy-lifting AI processing (Face Recognition) locally in the browser to ensure low-latency performance.
- **Server**: Manages data persistence, authentication, and administrative API endpoints.
- **Database**: A portable SQLite database stores student records, logs, and performance metrics.

6. Setup & Installation

Prerequisites

- Node.js (v18 or higher)
- npm or yarn

Steps

1. **Clone the repository**:

```
git clone <repository-url>
cd smart-attendance-system
```

2. **Install Dependencies**:

```
npm install
cd server && npm install
```

3. **Configure Environment**:
   - Create a `.env` file in the `server/` directory.
   - Set `PORT=8080` and `JWT_SECRET=your_secret`.
4. **Run the Application**:
   - Start Backend: `npm run dev:server` (from root)
   - Start Frontend: `npm run dev` (from root)
5. **Access**:
   - Open `http://localhost:5173` in your browser.

7. Future Enhancements

- **Multi-Camera Support**: Distributed attendance across large lecture halls.
- **Mobile Application**: Dedicated app for students to track their own attendance and performance.
- **Automatic Notifications**: Email/SMS alerts for low attendance or high performance.
- **Cloud Integration**: Migration to PostgreSQL for larger scale deployments.